

Evapo(transpi)ration

Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

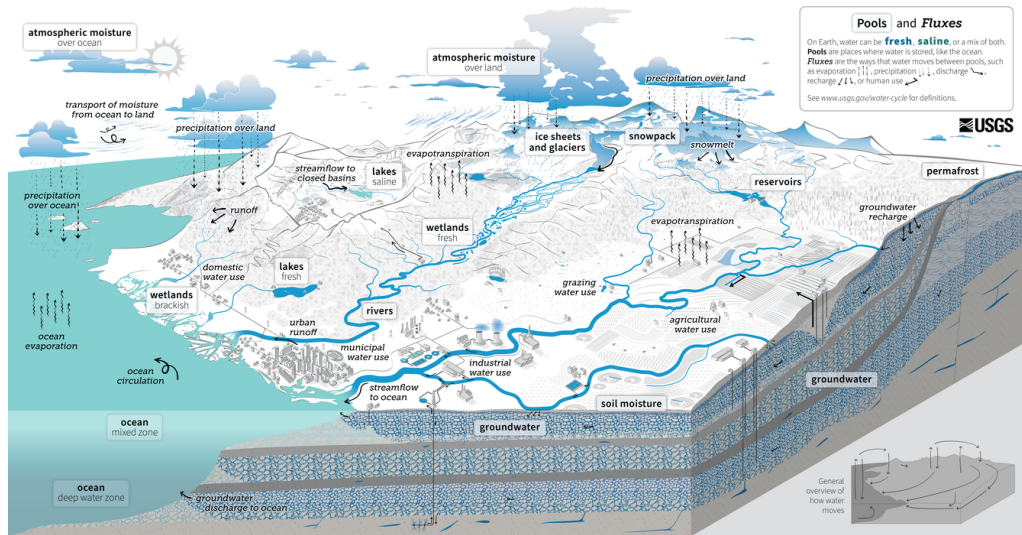
Housekeeping

- PS #4 assigned soon
- $\equiv$ is equivalence, not proportional/approximate, which is $\approx$

Learning Objectives

- 1 Evaporation processes
- 2 Methods for estimating potential evaporation
- 3 Transpiration process and estimation methods
- 4 Interception loss

Water Cycle



ET Classification

- 1 Type of surface
- 2 Water Availability
- 3 Stored energy use
- 4 Water-advected energy use

Potential Evap

open water
unlimited
it depends
it depends

Bare Soil

Atmosphere
controlled

soil

unlimited

x

x

water-
limited

soil

limited

x

x

transpi-
ration

leaves

limited

x

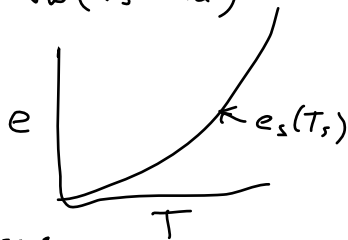
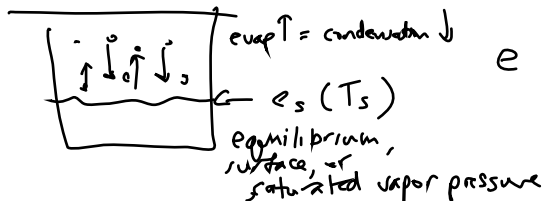
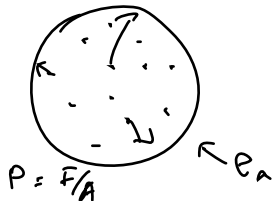
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Vapor Pressure

<https://www.youtube.com/watch?v=zmHaOWgNIDA>

- Take-Aways:
- Vapor pressure (e_a) is partial pressure of water vapor in a parcel of air
 - add vapor to parcel, e_a goes up
 - surface / saturation (e_s) vapor pressure is dependant on temperature
 - Evaporation is proportional to deficit between e_s + e_a (time wind and turbulence)
 - E goes up with T of liquid and wind and deficit

$$E \propto (e_s - a_a) \quad E = K_e \cdot V_w (e_s - a_a)$$



Free-water Evaporation in NM



Evaporation from reservoirs comprises 7% of total annual water use in NM

CA Aqueducts



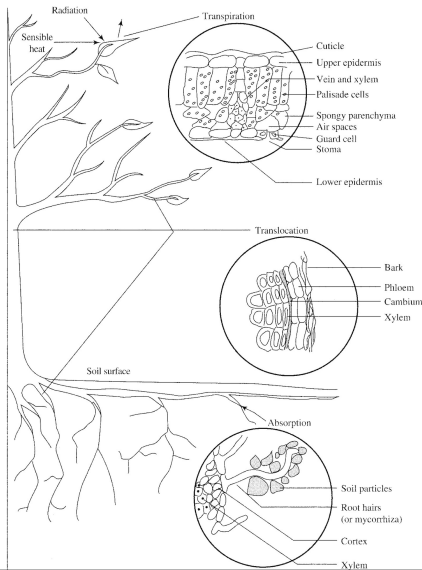
McKuin et al (2021) showed that covering all 4000 miles of aqueducts with solar panels could save 63 billion gallons of water each year or the annual residential water usage of 2 million people.

<https://www.sfgate.com/california/article/california-energy-experiment-solar.php>

Transpiration

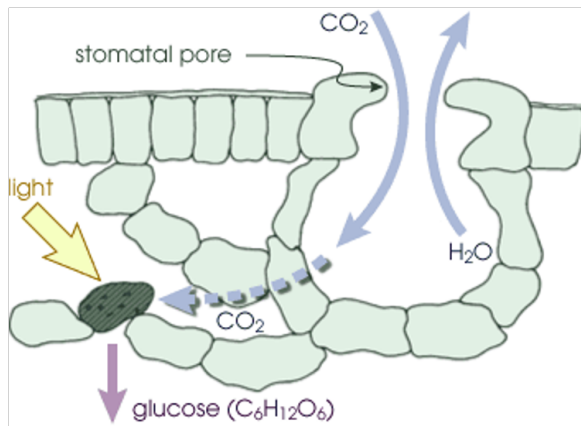
Transpiration is the evaporation of water from the vascular system of plants into the atmosphere.

Water is absorbed by the root system and translocated in liquid form through the vascular system of the roots, stems, branches, and leaves, until reaching the walls of stomatal cavities.



Photosynthesis

Transpiration occurs as a byproduct of photosynthesis. Plants open their stomatal pores (via guard cells, shown on the next slide) to dissolve atmospheric CO_2 into the water in the stomatal cavities. Since the vapor pressure is higher inside the leaf than the air outside, evaporation occurs.

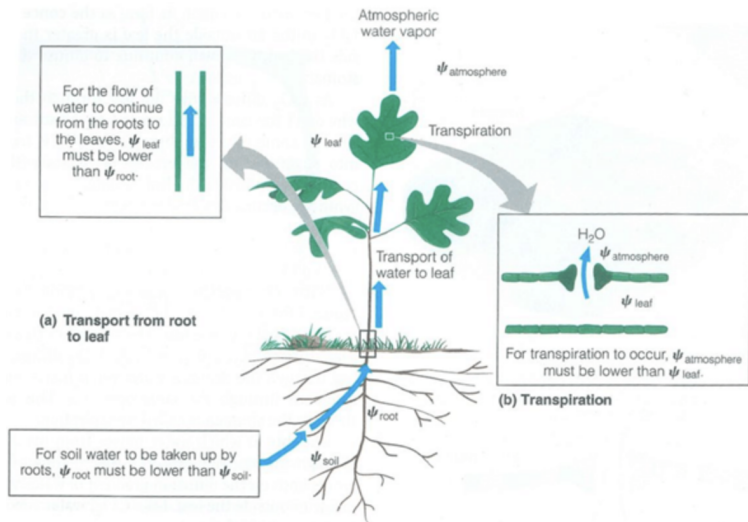


Stomata



Pressure Gradients

Water loss through stomatal pores creates pressure gradients that drive root water intake of soil moisture and translocation within xylem.



Estimating Transpiration

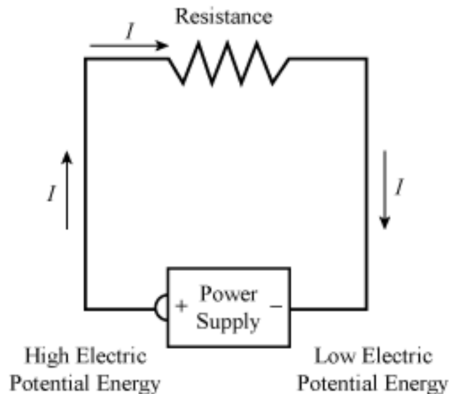
Essentially the same process as potential evaporation, but with an added component representing the extra resistance to evaporation from the size or stomatal openings.

Recall the mass-transfer equation for potential evaporation:

$$E = K_E \cdot v_a \cdot (e_s - e_a)$$

Detour to Ohm's Law

Electric current (I) is proportional to voltage (V) difference. The proportionality is governed by the resistance (R) of the conductor. **Conductance** is the inverse of resistance.



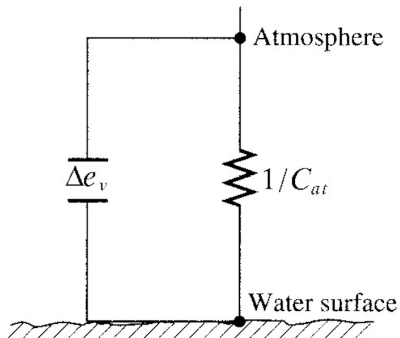
Estimating Transpiration

Recasting the mass transfer coefficient and wind speed term to atmospheric constant (K_a) and atmospheric conductance (C_{at}):

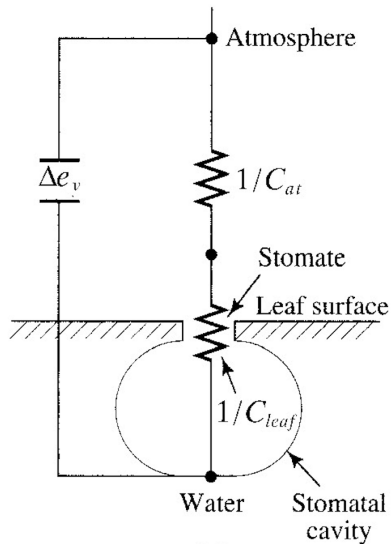
$$E = K_a \cdot C_{at} \cdot (e_s - e_a)$$

We do this because conductance is an easy way to scale up ET from leaf to an entire vegetated surface.

Evaporative Conductance



(a)



(b)

Leaf Conductance

Leaf conductance (C_{leaf}) is determined by the number of stomatal openings per unit area and size of those openings.

$$C_{leaf} = C_{leaf}^* \cdot f_K(K_{in}) \cdot f_\rho(\Delta\rho_v) \cdot f_T(T_a) \cdot f_\theta(\Delta\theta)$$

C_{leaf}^* is maximum leaf conductance K_{in}

incident SW radiation

$\Delta\rho_v$ is humidity deficit

T_a is air temp

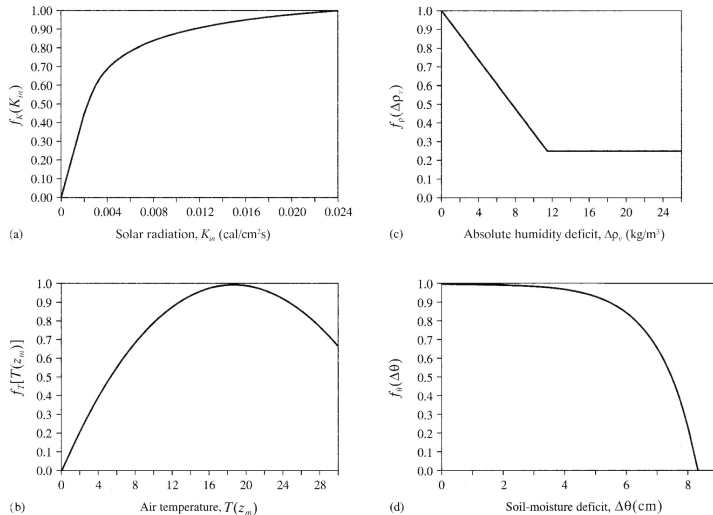
$\Delta\theta$ soil-moisture deficit

f are factors ranging from 0 to 1

Land Cover	C_{leaf}^* (mm s^{-1})
Conifer forest	5.3
Broadleaf forest	5.3
Savannah/shrub	5.3
Grassland	8.0
Tundra/nonforest wetland	6.6
Desert	5.0
Typical crop	11.0

Data from Federer et al. (1996).

Factors affecting Conductance

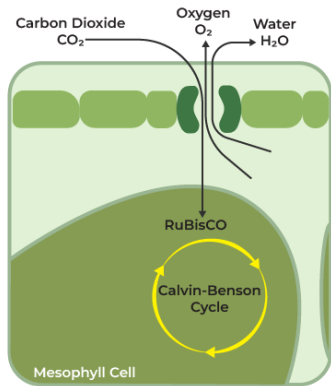


- Radiation (+)
- Absolute humidity deficit (-)
- Air temperature (+/-)
- Soil-moisture deficit (-)

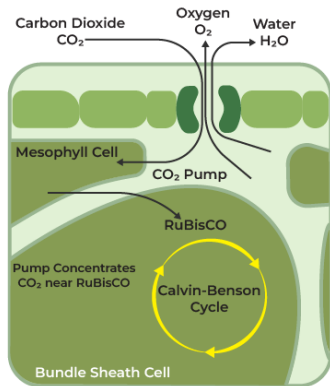
Figure 6.15 Effects of (a) solar radiation, K_m ; (b) air temperature, $T(z_m)$; (c) absolute-humidity deficit, Δp_v ; and (d) soil-moisture deficit, $\Delta\theta$ on relative leaf conductances (see table 6.5) [adapted from Stewart (1988)].

C3 and C4 plants

C4 plants typically have lower max. stomatal conductance compared to C3, and therefore have a higher water use efficiency (WUE), which is the ratio of carbon assimilated to water transpired.



C₃ Plant
Example: Elm



C₄ Plant
Example: Corn

Canopy Conductance

Like a circuit, you can scale up the number of resistors (one over conductance) in parallel by summing them. So, a uniform vegetated surface can be thought of as one big leaf with total conductance (C_{can}) proportional to sum of individual leaf conductances.

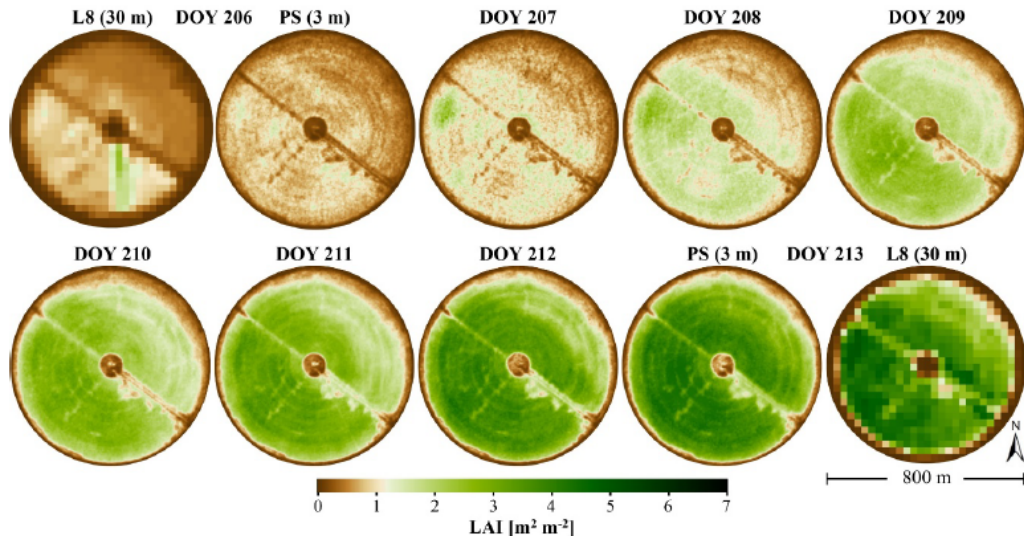
$$C_{can} = f_s \cdot LAI \cdot C_{leaf}$$

where LAI is the **leaf-area index** given by:

$$LAI = \frac{\text{total area of leaf-surface above area } A}{A} \quad (1)$$

and f_s is a shelter factor accounting for the fact that some leaves are sheltered from sun/wind and have lower ET rates.

Leaf Area Index (LAI)



Penman-Montieth Equation

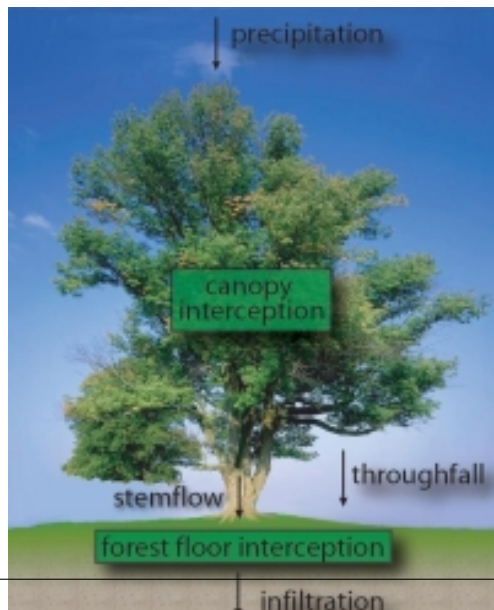
Recall the Penman model gave the best estimate of potential E that did not rely on difficult to measure components, such as T_s . Montieth (1965) showed that this equation could be modified to represent potential ET by incorporating canopy conductance.

$$ET = \frac{\Delta \cdot (K + L) + \rho_a \cdot c_a \cdot C_{at} \cdot e_a^* \cdot (1 - W_a)}{\rho_w \cdot \lambda_v \cdot [\Delta + \gamma \cdot (1 + \frac{C_{at}}{C_{can}})]} \quad (2)$$

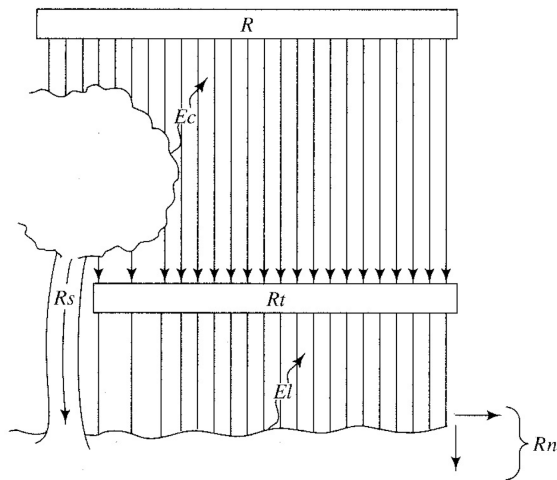
Like Penman, there is an assumption of no water-advected energy and no thermal storage, which may be violated in some cases (lakes, ugh)

Interception

Interception is the process by which precip lands on vegetative surfaces and is subject to evaporation (**interception loss**).



Interception



$$R_n = R - E_i \quad (3)$$

where:

- Net rainfall (R_n)
- Gross rainfall (R)
- Total interception loss (E_i)
- Throughfall (R_t)
- Stemflow (R_s)
- Canopy interception loss (E_c)
- Litter interception loss (E_L)

Interception Loss

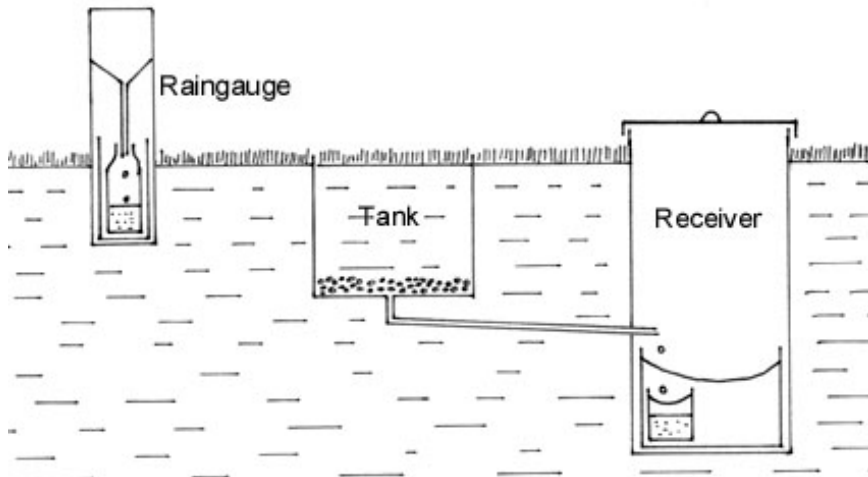
Interception loss can be estimated by measuring R , R_t , and R_s but has more commonly been done using empirical regression, analytical models, and physical/conceptual models, with estimates ranging from 10-40% in forested areas.

Annual Canopy Interception Loss as a Fraction of Gross Precipitation for Various Plant Communities				
Latitude	Location	Community	Annual Pptn. (cm) ^a	E_d/R
	Tropics	Lowland forest		0.22
	Tropics	Montane forest		0.18
3.0	Manaus, Brazil	Amazonian rain forest	281	0.09
4.0	Malaysia	Lowland forest		0.23
5.9	Ivory Coast	Evergreen hardwoods		0.09
6.5	W Java	Lowland tropical rain forest		0.21
7.0	Ghana	Semideciduous moist forest		0.16
10.0	Nigeria	Forest-savannah boundary		0.05
11.3	Kottamparamba, India	Cashew	300	0.31
18.3	Mts., E Puerto Rico	Tabonuco et al.	575	0.42
18.3	Mts., E Puerto Rico	Dwarf forest		0.09
34.5	Mts., A.R., U.S.	Pine-hardwood		0.13
35.0	W NC, U.S.	60-yr-old white pine	203	0.09
35.0	W NC, U.S.	Mixed hardwoods	203	0.12
35.0	W NC, U.S.	35-yr-old white pine	203	0.19
35.0	W NC, U.S.	10-yr-old white pine	203	0.15
42.2	S Is. New Zealand	Mixed evergreen hardwood	260	0.24
43.9	Mts., N NH, U.S.	Mixed hardwoods	130	0.13
-45.0	NW U.S.	Douglas fir		0.24
-45.0	NW U.S.	Douglas fir et al.		0.32
-45.0	NW U.S.	Sitka spruce-hemlock et al.		0.35
-45.0	NW U.S.	Mature Douglas fir		0.34
-45.0	NW U.S.	White pine-hemlock		0.21

Actual ET Estimation

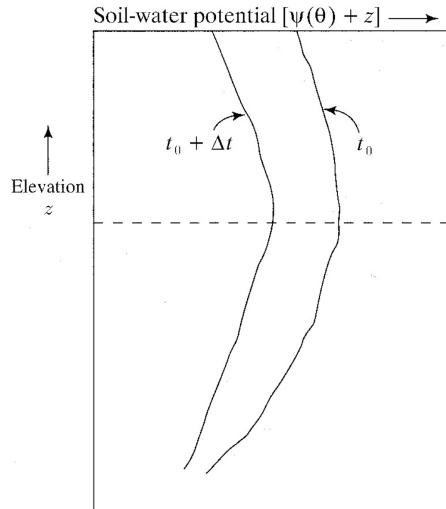
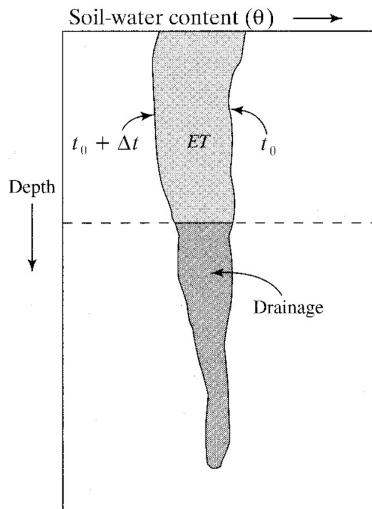
- 1 Penman-Montieth
- 2 Water Balance
- 3 Soil Moisture Balance
- 4 Lysimeter
- 5 Energy Balance
- 6 Eddy Correlation (more on this next week!)

Lysimeter



Soil Moisture Balance

"Lysimeter without walls"



ET Datasets

METRIC: Mapping Evapotranspiration at high Resolution with Internalized Calibration. Uses energy budget approach to estimate ET, relying on Landsat imagery (NDVI) and NLDAS re-analysis datasets for input

<https://eeflux-level1.appspot.com/>

OpenET uses an ensemble of satellite-based methods (including METRIC) to estimate AET. Bonus, it has an API that makes accessing the data via scripting a breeze!

<https://explore.etdata.org/#5/34.778/-103.931>